

1 Observation

- Attention maps show commas, periods and newlines receive disproportionately high scores.
- The hypothesis: a segment's information becomes condensed into the separator that closes it.
- This natural language structure can define data-dependent sparsity rather than a fixed pattern.

2 Fundamental design

THREE TOKEN SETS

- Every query attends to initial attention-sink tokens, a local window of neighboring tokens, and all separators.
- Other content tokens are masked in every head; a custom Sep-Attention kernel accelerates sparse multiplication.
- During generation, only KV states for these three groups are retained.

3 Streaming

FOUR-CACHE POLICY

- Maintain Initial, Separator, Past Window and Local Window caches under a fixed total capacity.
- When full, move separators from the past window into Separator Cache and discard other overflow tokens.
- Cache-relative positional shifting supports effectively unbounded streams.

4 Training

NATIVE, NOT POST-HOC

- SepLLM works training-free, from scratch or through post-training of a full-attention checkpoint.
- Training with the same sparse mask reduces the train-inference mismatch of cache eviction methods.
- It retains universal approximation ability; fixed-interval tokens cannot replace semantic separators.

5 Evidence

LESS KV, SIMILAR QUALITY

- Llama-3-8B training-free: 47.36% KV on GSM8K-CoT while maintaining 77.18% vs 77.79% accuracy.
- At equal loss, pretraining computation falls 28% and wall-clock training time 26%.
- Streaming works beyond 4M tokens; at 64K generation it uses 562 average KV vs 800 and lowers perplexity/time.

6 Meaning & limits

TAKEAWAY

- SepLLM treats punctuation-induced semantic units as a native sparse attention structure.
- It improves over StreamingLLM and generalizes across Pythia, Llama and Falcon families.
- Separator choices and cache sizes are hyperparameters; punctuation-poor data, code or other languages may require adaptation.

